

ELG 3106 Electromagnetic Engineering

Impact of Broadband Antireflection Coating Design on Solar
Power Production

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Summary

In this project, we will attempt to demonstrate how a given photovoltaic cell system will respond for different design constraints. Our system will constitute a solar cell covered with anti reflection coating of two or three layers. The constraints we will be changing are the number of layers, the refractive indices of each layer as well as their thickness.

By varying all these constraints, we hope to find the design approach which will maximize the power that is transmitted to the solar cell and thus minimize the reflection at the same time. A significant challenge is demonstrating that a reflective coating designed to nearly eliminate reflection at a specific wavelength may generate less power than a coating that moderately reduces reflection across a broader spectrum.

The use of an automation tool such as MATLAB will be highly useful as we will be testing and comparing results for thousands of different designs and numerically solving enormous amounts of calculations.

Considering the complexity of such a task, we will make a few assumptions that will help us lighten the calculations and the complexity of our mathematical approach to find the best layout for our antireflective coating. These assumptions will be outlined in the **Theory** section.

Lastly, we will employ the **Transfer Matrix Method (TMM)**, a powerful analytical tool for studying wave reflectivity across the layers of our system. Essentially, we will break down each layer of the system into a dynamical and propagation matrix which are functions dependent of the refractive indices of the different layers. With the multiplication of these matrices, we will be able to define one 'Global' reflection coefficient for the entire system which in turn will allow us to compute the reflectivity, transmissivity and other essential metrics for our analysis.

Introduction

Over the last decades, because of the emergence of many new technologies and humanity's burning desire to explore new horizons, the production of sustainable and reliable sources of energy is becoming a critical priority. While many forms of energy are already in use, issues such as global warming and pollution, are driving a shift towards greener alternatives.

One of these alternatives consists of harvesting the power from the sun. To do so we use a technology called photovoltaic cells. These special cells can capture the energy of light photons produced by the sun and extract electricity from them. A dozen years ago, this technology was extremely expensive, inefficient and did not have a long lifespan. Today, with significant breakthroughs in this field such as multi-junction Perovskite-coated silicon cells, it is expected that by 2030, solar cells alone will reach efficiencies of 30%¹ which is enormous compared to the peak efficiency of 15% in 2010.

An important factor affecting solar cell efficiency is the reflection of light off their surface which reduces the amount of energy absorbed. Luckily, scientist have studied different materials that have lower reflection for various wavelength of the solar light spectrum. The combination of very thin layers of such materials can create antireflection coatings that help transmit a maximum amount of energy to the solar cells.

In this report, we aim to determine the combinations of refractive indices of the materials forming the antireflective coating of double and triple-layer coatings that will allow us to obtain maximal efficiency. Using advance computational tools such as MATLAB will allow us to perform more complex and in-depth algorithms to find the optimal selection of materials to maximize solar cell performance.

¹ <https://www.cleanenergyreviews.info/blog/most-efficient-solar-panels>

Theory

General solar cell theory²

Solar cells function through a process called the photovoltaic effect, where incoming light particles (photons) will be absorbed by the electrons in the semi conductor material from which the solar cell is made. The energy that the photon transmits to the electron will “excite” it and give it enough energy to break free from the atomic bond it had with the semi conductor material. Doing so, it leaves behind an electron-hole pair (positive) which has the opposite charge of an electron (e^-). The solar cell is made with a p-n junction which is essentially a positively doped (p-type) semiconductor material (ex silicon and a small amount of boron) and a negatively doped (n-type) semiconductor material (silicon and small amount of phosphorous). It can also be made of other materials, but these are the most common ones. The p-n junction creates an electric field that will attract the freed electrons to the n-type and the positive holes to the p-type. Lastly, connecting tiny conducting wires from the n-type to a metal base plate under the p-type will allow the electrons to flow and create current when the sun shines on the cell.

Physics of our approach³

To determine the power produced by our solar cell, we will need to define a few variables. First, we have the idealized solar spectral irradiance $I(\lambda)$ that represents the wavelength distribution of light normally incident on the cell. It is characterized by the equation:

$$I(\lambda) = \frac{6.16 \times 10^{15}}{\lambda^5 (e^{2484/\lambda} - 1)}$$

Equation 1

Where λ is the wavelength (nm). This equation is idealized as it neglects the filtering effect of the earth's atmosphere to simplify the equation. To simplify our approach even further, we assume that the cell converts all photons incident on it into electricity with an efficiency of 100% for every wavelength.

² This section was written with inspiration from Dr Emad Gad's Electronics I course notes

³ This section was created with inspiration from Dr H. Schriemer's ELG3106 course notes and documents

Since the antireflective coating we are applying to the cell has a transmissivity $T(\lambda)$ we can calculate the electrical power generated by our cell with the following equation:

$$P = \int_{\lambda_1}^{\lambda_2} T(\lambda) I(\lambda) d\lambda$$

Equation 2

Where $\lambda_1 = 200\text{nm}$ and $\lambda_2 = 2200\text{nm}$, which represent the general range of the solar spectrum.

The Transfer Matrix Method TMM

An arbitrary multilayer system is shown in *Figure 1* below. In this figure we've annotated the electric field components traveling at each interface where the incident waves are traveling in the right (+) direction. The subscripts indicate the layer and the prime represents the waves on the right side of the interface

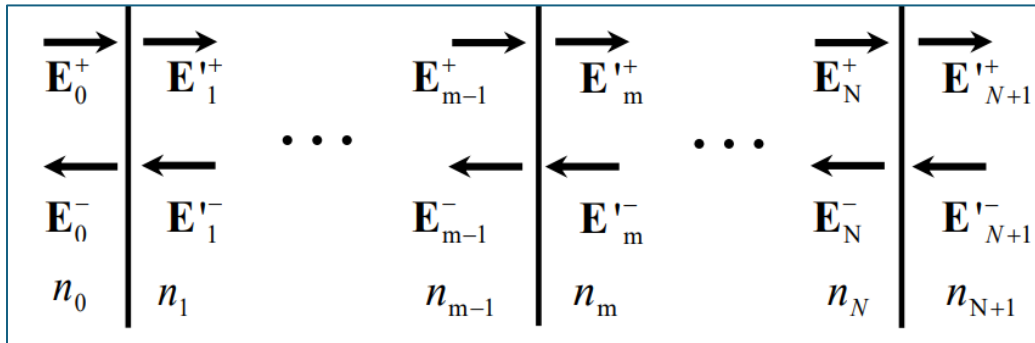


Figure 1: arbitrary N layer system

Each layer has a refractive index 'n'. The boundary conditions on the electric field vectors $\mathbf{E}$ on each side of the interfaces allow us to create a 2x2 matrix to describe the m^{th} interface. We can write this relation like so:

$$\begin{pmatrix} \mathbf{E}_{m-1}^+ \\ \mathbf{E}_{m-1}^- \end{pmatrix} = Q_{m-1,m} \begin{pmatrix} \mathbf{E}_m'^+ \\ \mathbf{E}_m'^- \end{pmatrix}$$

Where Q is the **dynamical matrix** defined in terms of reflection and transmission coefficients

$$Q_{m-1,m} = \frac{1}{\tau_{m-1,m}} \begin{bmatrix} 1 & \Gamma_{m-1,m} \\ \Gamma_{m-1,m} & 1 \end{bmatrix}$$

Since we are only considering the normally incident waves, these coefficients can be calculated as

$$\Gamma_{m-1,m} = \frac{n_{m-1} - n_m}{n_{m-1} + n_m} \quad \text{and} \quad \tau_{m-1,m} = \frac{2n_{m-1}}{n_{m-1} + n_m}$$

The field components on each side of the m^{th} layer will accumulate a phase difference when travelling between 2 media of different refraction indices. These field components will be related by the **propagation matrix P**

$$\begin{pmatrix} \mathbf{E}_m'^+ \\ \mathbf{E}_m'^- \end{pmatrix} = P_m \begin{pmatrix} \mathbf{E}_m^+ \\ \mathbf{E}_m^- \end{pmatrix}$$

where

$$P_m = \begin{bmatrix} \exp(j\delta_m) & 0 \\ 0 & \exp(-j\delta_m) \end{bmatrix}$$

and $\delta_m = \frac{2\pi}{\lambda_0} * n_m * d_m$ is the phase ‘thickness’ of the m^{th} layer and d_m is the physical thickness of the layer. For a system comprised of N layers, finding the final transmitted electric field will require the application of the dynamical and propagation matrices N+1 times in succession. The product of all these 2x2 matrices is the transfer matrix of the system (**T**).

$$\begin{pmatrix} \mathbf{E}_0^+ \\ \mathbf{E}_0^- \end{pmatrix} = \mathbf{T} \begin{pmatrix} \mathbf{E}_{N+1}'^+ \\ \mathbf{E}_{N+1}'^- \end{pmatrix}$$

The Transfer matrix can be written in a condensed form as

$$\mathbf{T} = \begin{bmatrix} T_{1,1} & T_{1,2} \\ T_{2,1} & T_{2,2} \end{bmatrix} = Q_{0,1} \prod_{m=1}^N P_m Q_{m,m+1}$$

By setting $\mathbf{E}_i = \mathbf{E}_0^+$, $\mathbf{E}_r = \mathbf{E}_0^-$, $\mathbf{E}_{tr} = \mathbf{E}_{N+1}^+$ We can define Γ and τ in terms of the $\mathbf{T}$ matrix components.

$$\Gamma = \frac{\mathbf{E}_r}{\mathbf{E}_i} = \frac{T_{2,1}}{T_{1,1}} \quad \text{and} \quad \tau = \frac{\mathbf{E}_t}{\mathbf{E}_i} = \frac{1}{T_{1,1}}$$

Finally, for a lossless, nonmagnetic medium, we can relate the reflectivity and transmissivity coefficients with this simple equation

$$|\Gamma|^2 + |\tau|^2 \left(\frac{n_{N+1}}{n_0} \right) = 1$$

This last equation can be simplified using Refelctivity (R) and Transmissivity (T)

$$R + T = 1$$

Design Approach

For our design, we will only focus on the double and triple layer antireflection coatings. To simplify our problem, we will assume that the sunlight is modelled as a uniform plane wave (UPW) and that it is normally incident to the coating and the cell, meaning all oblique incident waves are disregarded. We also assume the incident wave is travelling in air and that the solar cell is made of silicone with a refractive index of 3.5.

Each layer of the coating will have a thickness of one-quarter of the centre wavelength in the material. We presume that the centre wavelength in air is $\lambda_c = 650\text{nm}$. Thus, we can find the wavelength in each material using the equation $\lambda_m = \frac{\lambda_0}{n_m}$. Therefore, the thickness $d_m = \lambda_m/4$

Between each layer, an accumulation phase will appear for all the different wave lengths. This phase is given by $\delta_m = \frac{2\pi}{\lambda_0} * n_m * d_m$

From these values, we can build the TM using MATLAB that will help us compute the reflection coefficient of the system and then we will create algorithms to test thousands of different refractive indices for each layer to pinpoint the best solutions.

The double layer coating Part 1

A) Finding the reflectivity at the center wavelength for a system with no anti reflective coating

To find the reflectivity of our cell, we first need to find its reflection coefficient. Since we assume that the waves were normally incident to the surface, our reflection coefficient can be calculated as

$$\Gamma_{m-1,m} = \frac{n_{m-1} - n_m}{n_{m-1} + n_m}$$

Where n_{m-1} is the refractive index of air (1), and n_m is the refractive index of our cell (3.5). By substituting our values, we can find

$$\Gamma_{air,cell} = \frac{1 - 3.5}{1 + 3.5} = -0.556$$

The reflectivity is simply the square of the magnitude of the reflection coefficient

$$R = |\Gamma_{air,cell}|^2 = (-0.556)^2 = 0.3086$$

We can calculate the power that will be transmitted to the cell using *Equation 1*. To find this value, we will also need to know the Transmissivity which is directly related to the Reflectivity as

$$R + T = 1 \rightarrow T = 1 - R = 1 - 0.3086 = 0.6914$$

Therefore, the power at the center frequency $\lambda_c = 650\text{nm}$ is

$$P = \frac{0.6914 * 6.16 * 10^{15}}{650^5 (e^{\frac{2484}{650}} - 1)} = 0.8216 \left(\frac{W}{m^2} \right)$$

B) Applying the TMM method

For this second part, we will attempt to determine the values of the reflection and transmission coefficients using the **TMM method** and the equations defined in the **Theory** part. Finding these values for a 2-layer antireflection coating will allow us to determine the power that is transferred to the cell with *Equation 1 and 2*.

For this we begin by breaking apart the system consisting of these 2 layers

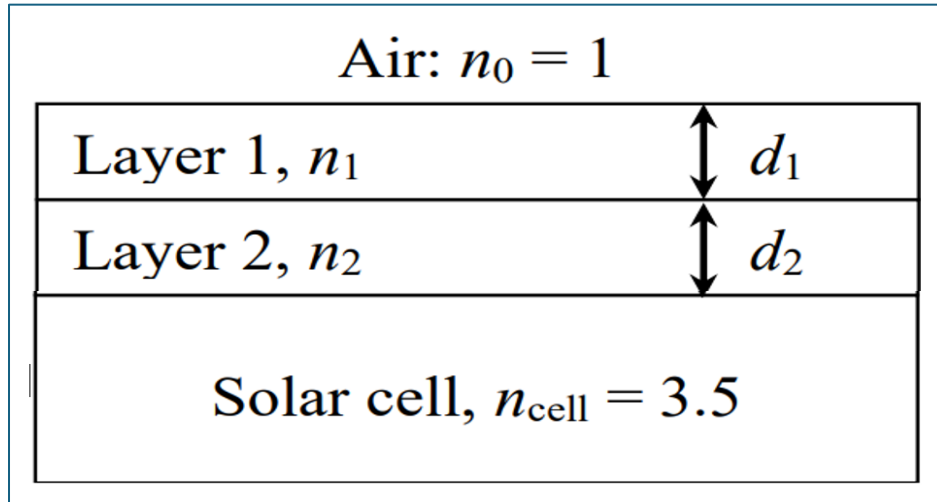


Figure 1 Solar cell system layout representing refractive indices

We know the electric field will propagate both ways between each layer as seen in *Figure 1*. So, the transfer matrix equation would be

$$T = Q_{0,1}P_1Q_{1,2}P_2Q_{2,3}$$

We've seen in the theory that the phase thickness $\delta_m = \frac{2\pi}{\lambda_0} * n_m * d_m$. Considering d_m is $\lambda_0/4n_m$. Which leads to $\delta_m = \frac{\pi}{2}$ so the propagation matrix therefore becomes

$$P_m = \begin{bmatrix} e^{j*\delta_m} & 0 \\ 0 & e^{-j*\delta_m} \end{bmatrix} = j * \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

Similarly, we can calculate the three dynamical matrices Q_{01} Q_{12} and Q_{23} as

$$\frac{1}{\tau_{01}} * \begin{bmatrix} 1 & \Gamma_{01} \\ \Gamma_{01} & 1 \end{bmatrix}, \quad \frac{1}{\tau_{12}} * \begin{bmatrix} 1 & \Gamma_{12} \\ \Gamma_{12} & 1 \end{bmatrix} \text{ and } \quad \frac{1}{\tau_{23}} * \begin{bmatrix} 1 & \Gamma_{23} \\ \Gamma_{23} & 1 \end{bmatrix}, \text{ respectively.}$$

Knowing all the values of the propagation and dynamical matrices will now allow us to construct the transfer matrix **T** in MATLAB and compute any values such as the refractive indices at the center wavelength.

C) Finding the reflectivity as a function of the refractive indices analytically.

Once again considering a cell with a double layer coating, we can define the reflectivity in terms of the refractive indices by solving the transfer matrix method we used in part b).

Expanding to the matrix form we get

$$T = \frac{j*j}{\tau_{01}\tau_{12}\tau_{23}} * \begin{bmatrix} 1 & \Gamma_{01} \\ \Gamma_{01} & 1 \end{bmatrix} * \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} * \begin{bmatrix} 1 & \Gamma_{12} \\ \Gamma_{12} & 1 \end{bmatrix} * \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} * \begin{bmatrix} 1 & \Gamma_{23} \\ \Gamma_{23} & 1 \end{bmatrix}$$

We can remove the constant $\frac{j*j}{\tau_{01}\tau_{12}\tau_{23}}$ from our equation because it will cancel out later

$$T = \begin{bmatrix} 1 - \Gamma_{01}\Gamma_{12} & \Gamma_{01} - \Gamma_{12} \\ \Gamma_{01} - \Gamma_{12} & 1 - \Gamma_{01}\Gamma_{12} \end{bmatrix} * \begin{bmatrix} 1 & \Gamma_{23} \\ \Gamma_{23} & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 1 - \Gamma_{01}\Gamma_{12} + [\Gamma_{01} - \Gamma_{12}]\Gamma_{23} & (1 - \Gamma_{01}\Gamma_{12})\Gamma_{23} + \Gamma_{01} - \Gamma_{12} \\ \Gamma_{01} - \Gamma_{12} + (1 - \Gamma_{01}\Gamma_{12})\Gamma_{23} & (\Gamma_{01} - \Gamma_{12})\Gamma_{23} + 1 - \Gamma_{01}\Gamma_{12} \end{bmatrix}$$

From our previous equations, we know that $\Gamma = \frac{T_{2,1}}{T_{1,1}} = \frac{\Gamma_{01} - \Gamma_{12} + (1 - \Gamma_{01}\Gamma_{12})\Gamma_{23}}{1 - \Gamma_{01}\Gamma_{12} + [\Gamma_{01} - \Gamma_{12}]\Gamma_{23}}$

We can further develop this equation to write it in terms of the refractive indices

$$\Gamma = \frac{T_{2,1}}{T_{1,1}} = \frac{\left(\frac{n_0 - n_1}{n_0 + n_1}\right) - \left(\frac{n_1 - n_2}{n_1 + n_2}\right) + \left(\frac{n_2 - n_3}{n_2 + n_3}\right) - \left(\frac{n_0 - n_1}{n_0 + n_1}\right) \left(\frac{n_1 - n_2}{n_1 + n_2}\right) \left(\frac{n_2 - n_3}{n_2 + n_3}\right)}{1 - \left(\frac{n_0 - n_1}{n_0 + n_1}\right) \left(\frac{n_1 - n_2}{n_1 + n_2}\right) + \left(\frac{n_0 - n_1}{n_0 + n_1}\right) \left(\frac{n_2 - n_3}{n_2 + n_3}\right) - \left(\frac{n_1 - n_2}{n_1 + n_2}\right) \left(\frac{n_2 - n_3}{n_2 + n_3}\right)}$$

Lastly, we know that the reflectivity is none other than the square of our reflection coefficient

$$R = |\Gamma|^2 = \left(\frac{\left(\frac{n_0 - n_1}{n_0 + n_1}\right) - \left(\frac{n_1 - n_2}{n_1 + n_2}\right) + \left(\frac{n_2 - n_3}{n_2 + n_3}\right) - \left(\frac{n_0 - n_1}{n_0 + n_1}\right) \left(\frac{n_1 - n_2}{n_1 + n_2}\right) \left(\frac{n_2 - n_3}{n_2 + n_3}\right)}{1 - \left(\frac{n_0 - n_1}{n_0 + n_1}\right) \left(\frac{n_1 - n_2}{n_1 + n_2}\right) + \left(\frac{n_0 - n_1}{n_0 + n_1}\right) \left(\frac{n_2 - n_3}{n_2 + n_3}\right) - \left(\frac{n_1 - n_2}{n_1 + n_2}\right) \left(\frac{n_2 - n_3}{n_2 + n_3}\right)} \right)^2$$

D) Minimizing the reflectivity in terms of the refractive indices when $n_1=1.4$

Because we are trying to find the equation in terms of the refractive indices that minimizes the **reflectivity** at the center wavelength, we calculate

$$\Gamma = \frac{T_{2,1}}{T_{1,1}} = 0 \rightarrow T_{2,1} = 0$$

This give us the following equation to solve:

$$\Gamma_{01} - \Gamma_{12} + (1 - \Gamma_{01}\Gamma_{12})\Gamma_{23} = 0$$

$$\left(\frac{n_0 - n_1}{n_0 + n_1}\right) - \left(\frac{n_1 - n_2}{n_1 + n_2}\right) + \left(\frac{n_2 - n_3}{n_2 + n_3}\right) - \left(\frac{n_0 - n_1}{n_0 + n_1}\right) \left(\frac{n_1 - n_2}{n_1 + n_2}\right) \left(\frac{n_2 - n_3}{n_2 + n_3}\right) = 0$$

$$2 * n_0 n_2 (n_2 + n_3) - 2 * (n_1)^2 (n_2 + n_3) + 2 * n_0 n_2 (n_2 - n_3) + 2 * (n_1)^2 (n_2 - n_3) = 0$$

$$n_0(n_2)^2 = (n_1)^2 n_3$$

In this equation, n_0 is the refractive index of air which is 1, $n_1=1.4$ and $n_3=n_{cell}=3.5$. We can find the optimal choice for n_2 that will minimize the reflectivity by solving for n_2 in our previous equation

$$n_2 = n_1 * \sqrt{\frac{n_{cell}}{n_0}} = 1.4 * \sqrt{3.5} = 2.62$$

The refractive index for the second layer of the antireflection coating must be 2.62 in order to minimize the Reflectivity if our first layer has an index of 1.4

The double layer coating Part 2

A) Graphing the total reflectivity spectrum as a function of wavelength in free space from 400 to 1400nm

For this part, we assume that the values of the refractive indices are the ones we found in **part 1 d)**. This means $n_0=1$, $n_1=1.4$, $n_2=2.62$ and $n_{\text{cell}}=3.5$. We can now find the value for the reflectivity in MATLAB by changing the wavelength in the propagation matrix but keeping the thickness constant which gives us

$$P_m = \begin{bmatrix} e^{j\frac{\pi}{2}\frac{\lambda_c}{\lambda}} & 0 \\ 0 & e^{-j\frac{\pi}{2}\frac{\lambda_0}{\lambda}} \end{bmatrix}$$

Calculating the values in MATLAB allowed us to trace the following graph:

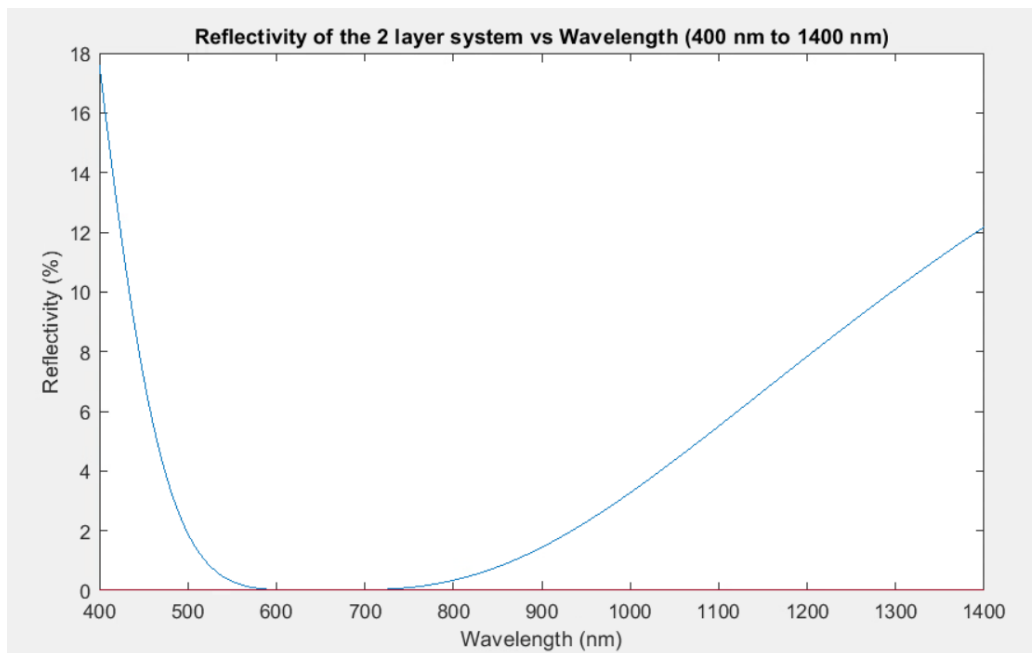


Figure 2 Reflectivity vs the wavelength for the 2-layer antireflection coating

As we clearly see from this graph, the reflectivity is minimized around a value of 650 nm which confirms that our calculations in part 1 c) and d) were correct and that when $n_2=2.62$, we are in fact optimizing the effectiveness of our coating at the center wavelength.

B) Integrating the spectrum to find the transmitted power to the solar cell.

While trying to find the power transmitted to the cell, we needed to add a few parts to our MATLAB code. We must now calculate the Transmissivity and the Irradiance value to apply *Equation 2* to find the power. Since the computed values are discrete, meaning each values for the wavelength are integers, we can calculate this integral by finding the sum of all the values of the integrand $T(\lambda)*I(\lambda)$ for all values within the integration limits.

Integration bounds of $\lambda_1=400\text{nm}$ and $\lambda_2= 1400\text{nm}$

Doing so, we find that the total transmitted power is 751.2021 Watts, for a surface of 1m^2 which is the area under the curve of the power versus the wavelength that we can plot

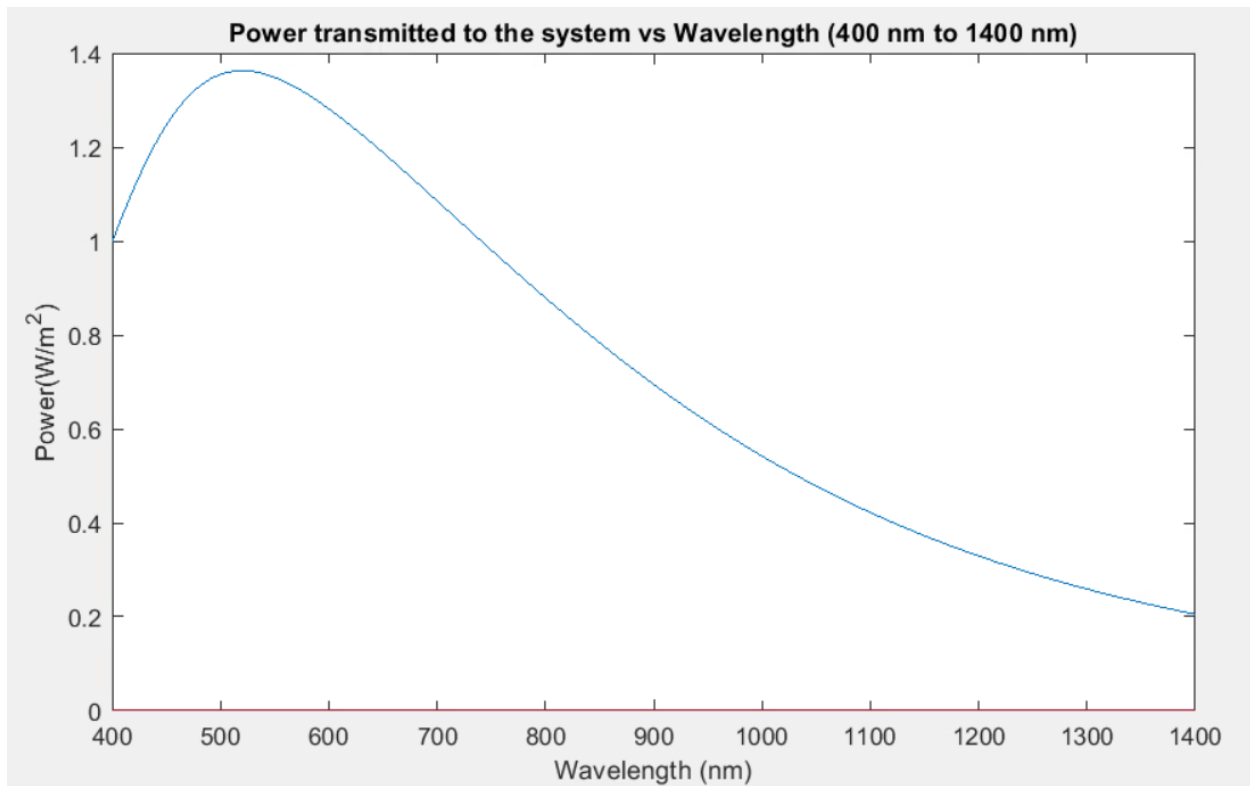


Figure 3 Graph of the Power versus the Wavelength from (400 to 1400 nm)

Integration bounds of $\lambda_1=200\text{nm}$ and $\lambda_2= 2200\text{nm}$

By applying the same steps as the previous range, we can easily find using our MATLAB code that the total power transmitted to the cell for an area of 1m^2 is 929.5916 Watts. Similarly, we can also obtain the graph of the power versus the wavelength for this new range.

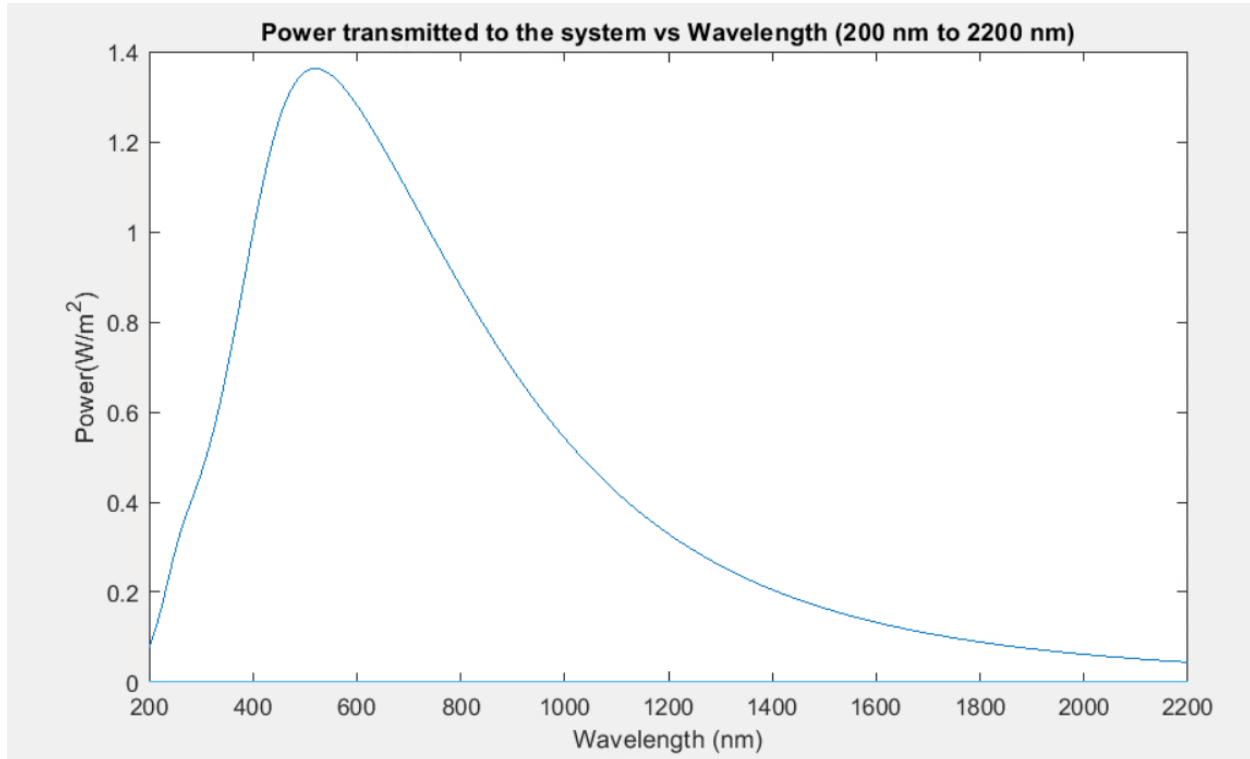


Figure 4 Graph of the Power transmitted vs the wavelength (200 to 2200nm)

As expected, the maximal power is obtained around the wavelengths where the Reflectivity in **part 2 a)** was at its lowest for both spectrum ranges.

C) Optimizing the power transmission by finding ideal refractive indices for the layers

Increasing the power that will be transmitted to the cell is possible by modifying the refractive indices and thickness of each layer of the anti reflection coating. This means that even if we've previously optimized the value of n_2 in order to completely eliminate the reflectivity at the center wavelength, there are other values that might not fully eliminate the reflection at that specific wavelength but when you consider a broader spectrum, these parameter will be more efficient and produce more energy.

By running the optimization algorithm that calculated the total power for each combination of refractive indices for n_1 and n_2 between 1 and 4 in increments of 0.01, we analyzed a total of 90 000 possibilities and discovered these results :

```
>> Project_2_c_optimization  
The max power obtainable is: 758.887624  
for n1 = 1.460000 and n2 = 2.410000
```

So for a value of $n_1=1.46$ and $n_2=2.41$ we can attain a power of 758.8876 Watts/m². Which is superior to the previous results in **part 2b**).

It is possible to plot the Reflectivity vs the wavelength for the optimal selection of our refractive indices which does not minimize the reflectivity at the center wavelength but rather it minimizes it at other wavelengths.

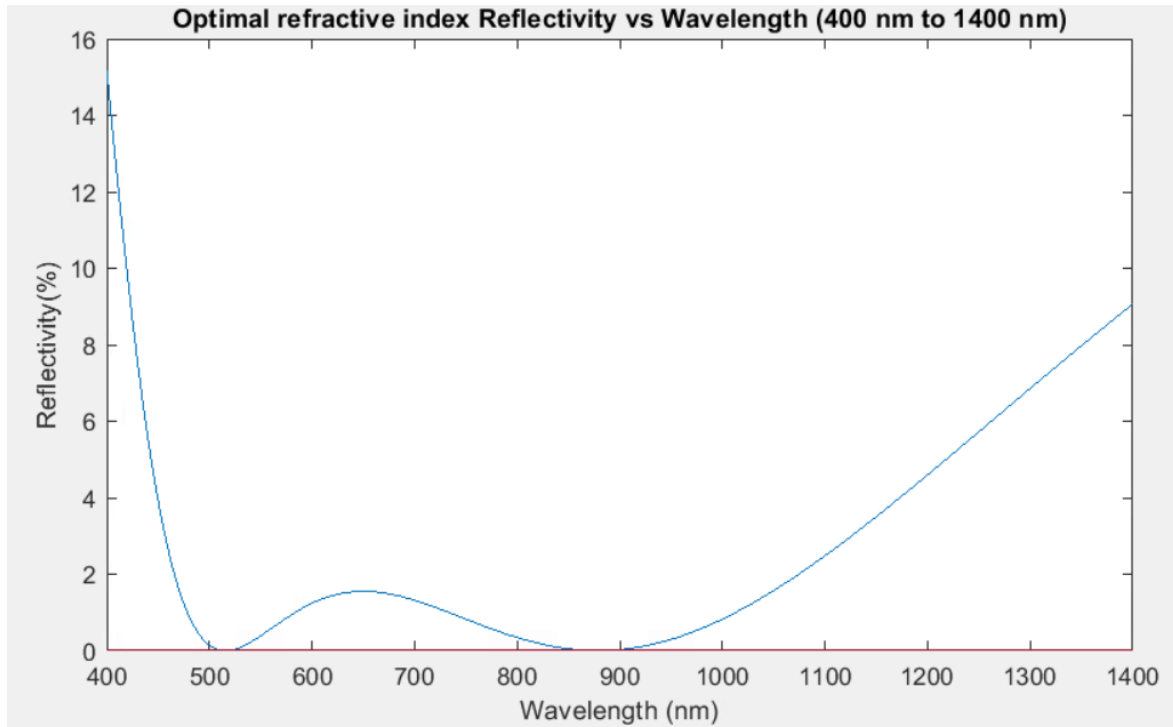


Figure 5 Graph of the Reflectivity versus wavelength for optimal power transfer in the range 400nm to 1400nm for $n_1=1.46$ and $n_2=2.41$

We can compare the spectrum optimized reflectivity graph with our center wavelength optimized graph to get a better understanding of our results:

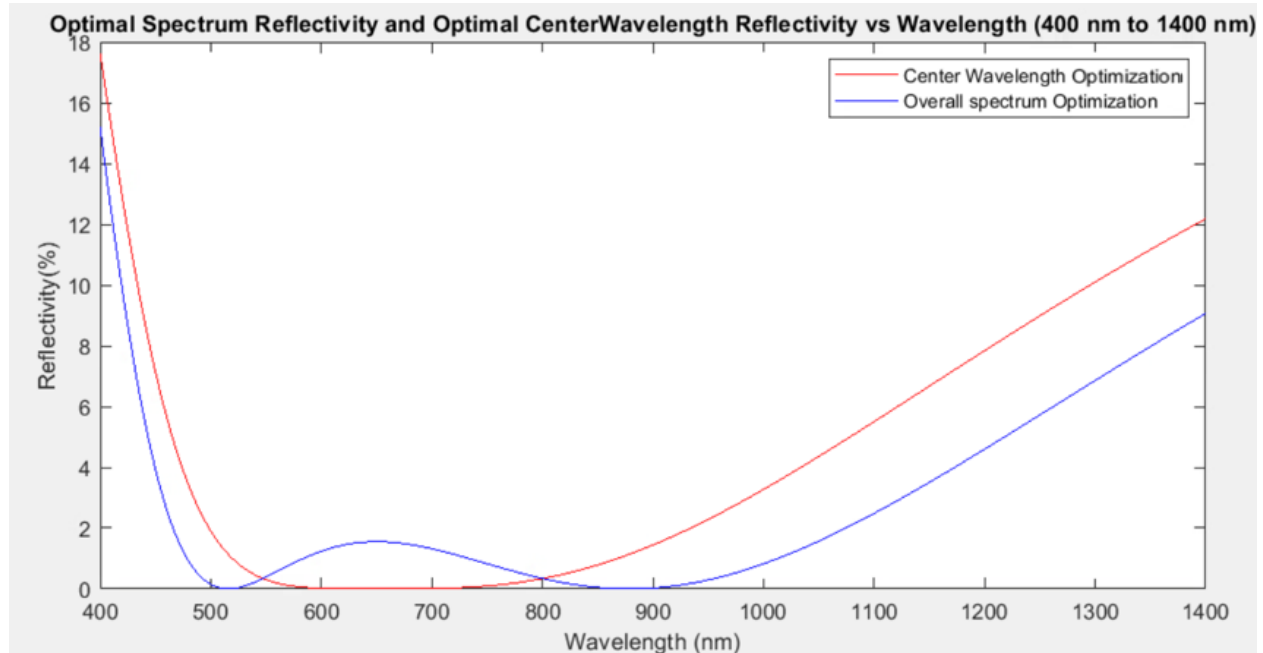


Figure 6 Graph of optimized Reflectivity over the entire spectrum and Optimized reflectivity at the center wavelength vs Wavelength (400 nm to 1400 nm)

As We see in this graph, the method that only optimizes the Reflectivity at the **center wavelength** has a better **Transmissivity** at this wavelength than the system that optimizes the Reflectivity across a broader spectrum. Yet, the overall power transmission is lower for the red curve. Its easy to deduce this by observing the two curves where we can see that the blue one has a much sharper drop for lower wavelengths and that for wavelengths superior to 900 it remains inferior to red one by around 3%.

If we repeat this experiment with a broader range from 200nm to 2200nm, we obtain different values maximizing our results :

```
>> Project_2_c_optimization  
The max power obtainable is: 941.414284  
for n1 = 1.480000 and n2 = 2.380000
```

For optimized antireflection coating layer values $n_1=1.48$ and $n_2=2.38$, we can obtain a power of 941.414W/m².

We can plot the reflectivity of this optimized parameter selection

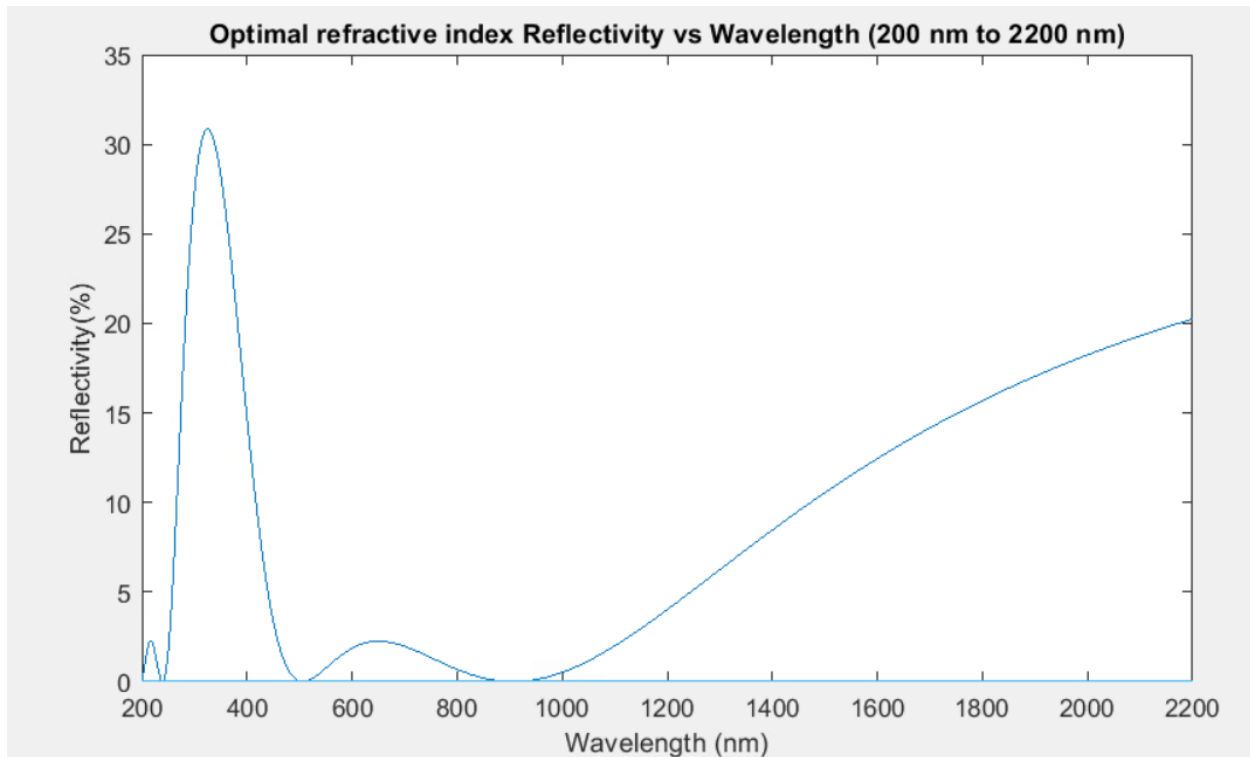


Figure 7 Graph of the Reflectivity versus wavelength (200 to 2200nm) for optimized n_1 and n_2 values

We can also compare our two graph to visualize the differences in our optimization methods.

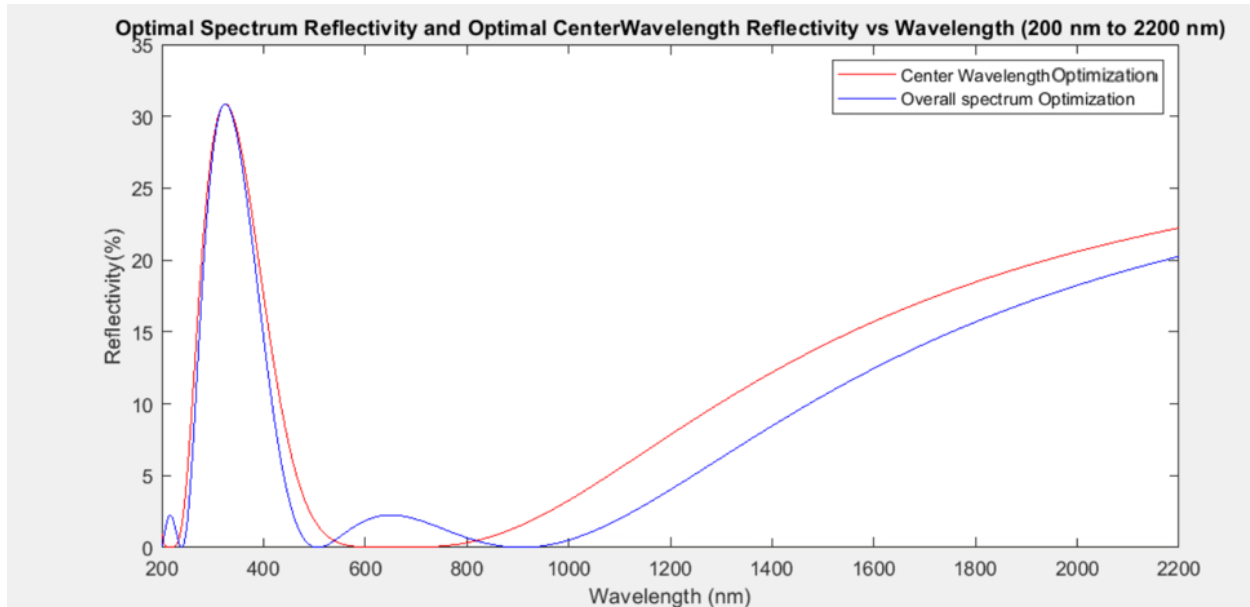


Figure 8 Graph of optimized Reflectivity over the entire spectrum and Optimized reflectivity at the center wavelength vs Wavelength (200 nm to 2200 nm)

Once again, we can see that the center wavelength optimization will have a lower reflectivity and higher power for wavelengths around 650 but for the rest of the wavelengths, the reflectivity will be lower for the selection of n_1 and n_2 that optimizes the total power. This is why the peak around 300-400nm is also sharper for the full spectrum optimization method and that for wavelengths above 900, the reflectivity for the blue curve is relatively lower than the red curve.

The Triple-Layer Coating Part 3

A) Analytically finding the reflectivity at the central wavelength in terms of the refraction indices

In a very similar fashion to **part 1 c)**, we will first start by defining the Transfer Matrix that will allow us to find the reflectivity. For a triple-layer system, we will require three propagation matrices and four dynamical ones. Our system is shown in *Figure 9* and our incident waves will need to travel all three layers before reaching the solar cell.

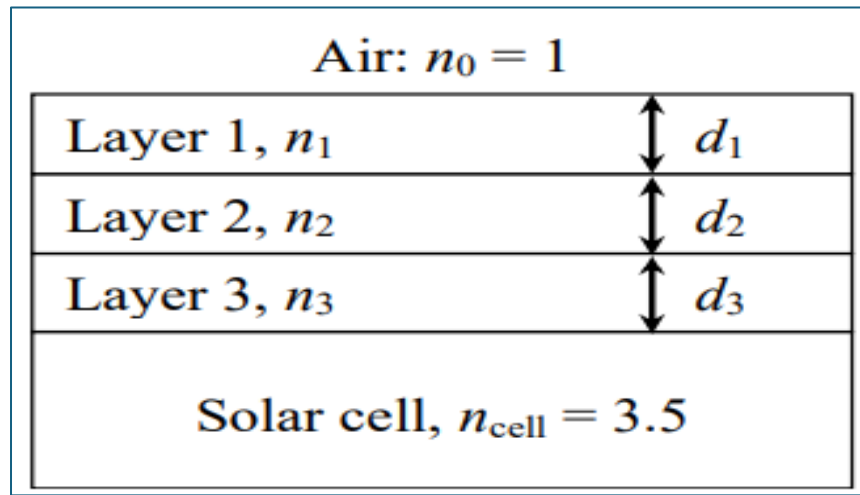


Figure 9 Triple layer anti-reflective coating system

After considering the nature of our three layer system, we are ready to build our TM.

This gives us the following equation

$$T = Q_{0,1}P_1Q_{1,2}P_2Q_{2,3}P_3Q_{3,4}$$

Just like the double-layer example, we can define our matrices as

$$P_m = \begin{bmatrix} e^{j\delta_m} & 0 \\ 0 & e^{-j\delta_m} \end{bmatrix} = j * \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

Similarly, we can calculate the four dynamical matrices Q_{01} , Q_{12} , Q_{23} and Q_{34} , respectively as

$$\frac{1}{\tau_{01}} * \begin{bmatrix} 1 & \Gamma_{01} \\ \Gamma_{01} & 1 \end{bmatrix}, \quad \frac{1}{\tau_{12}} * \begin{bmatrix} 1 & \Gamma_{12} \\ \Gamma_{12} & 1 \end{bmatrix}, \quad \frac{1}{\tau_{23}} * \begin{bmatrix} 1 & \Gamma_{23} \\ \Gamma_{23} & 1 \end{bmatrix} \text{ and } \frac{1}{\tau_{34}} * \begin{bmatrix} 1 & \Gamma_{34} \\ \Gamma_{34} & 1 \end{bmatrix}$$

Combining these together we get

$$T = \frac{j * j * j}{\tau_{01} \tau_{12} \tau_{23} \tau_{34}} * \begin{bmatrix} 1 & \Gamma_{01} \\ \Gamma_{01} & 1 \end{bmatrix} * \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} * \begin{bmatrix} 1 & \Gamma_{12} \\ \Gamma_{12} & 1 \end{bmatrix} * \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} * \begin{bmatrix} 1 & \Gamma_{23} \\ \Gamma_{23} & 1 \end{bmatrix} * \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} * \begin{bmatrix} 1 & \Gamma_{34} \\ \Gamma_{34} & 1 \end{bmatrix}$$

We can ignore $\frac{j * j * j}{\tau_{01} \tau_{12} \tau_{23} \tau_{34}}$ because it will not impact the result of the reflection coefficient since it will cancel out in the division.

Multiplying the matrices will give us

$$T = \begin{bmatrix} 1 - \Gamma_{01} \Gamma_{12} & \Gamma_{01} - \Gamma_{12} \\ \Gamma_{01} - \Gamma_{12} & 1 - \Gamma_{01} \Gamma_{12} \end{bmatrix} * \begin{bmatrix} 1 - \Gamma_{23} \Gamma_{34} & \Gamma_{34} - \Gamma_{23} \\ \Gamma_{23} - \Gamma_{34} & \Gamma_{23} \Gamma_{34} - 1 \end{bmatrix}$$

$$= \begin{bmatrix} (1 - \Gamma_{01} \Gamma_{12})(1 - \Gamma_{23} \Gamma_{34}) + (\Gamma_{01} - \Gamma_{12})(\Gamma_{23} - \Gamma_{34}) & (1 - \Gamma_{01} \Gamma_{12})(\Gamma_{34} - \Gamma_{23}) + (\Gamma_{01} - \Gamma_{12})(\Gamma_{23} \Gamma_{34} - 1) \\ (\Gamma_{01} - \Gamma_{12})(1 - \Gamma_{23} \Gamma_{34}) + (1 - \Gamma_{01} \Gamma_{12})(\Gamma_{23} - \Gamma_{34}) & (\Gamma_{01} - \Gamma_{12})(\Gamma_{34} - \Gamma_{23}) + (1 - \Gamma_{01} \Gamma_{12})(\Gamma_{23} \Gamma_{34} - 1) \end{bmatrix}$$

We can calculate the reflection coefficient as $\Gamma = \frac{T_{2,1}}{T_{1,1}}$

$$\Gamma = \frac{T_{2,1}}{T_{1,1}} = \frac{(\Gamma_{01} - \Gamma_{12})(1 - \Gamma_{23} \Gamma_{34}) + (1 - \Gamma_{01} \Gamma_{12})(\Gamma_{23} - \Gamma_{34})}{(1 - \Gamma_{01} \Gamma_{12})(1 - \Gamma_{23} \Gamma_{34}) + (\Gamma_{01} - \Gamma_{12})(\Gamma_{23} - \Gamma_{34})}$$

$$\Gamma = \frac{\left[\left(\frac{n_0 - n_1}{n_0 + n_1} \right) - \left(\frac{n_1 - n_2}{n_1 + n_2} \right) \right] \left[1 - \left(\frac{n_2 - n_3}{n_2 + n_3} \right) \left(\frac{n_3 - n_4}{n_3 + n_4} \right) \right] + \left[1 - \left(\frac{n_0 - n_1}{n_0 + n_1} \right) \left(\frac{n_1 - n_2}{n_1 + n_2} \right) \right] \left[\left(\frac{n_2 - n_3}{n_2 + n_3} \right) - \left(\frac{n_3 - n_4}{n_3 + n_4} \right) \right]}{\left[1 - \left(\frac{n_0 - n_1}{n_0 + n_1} \right) \left(\frac{n_1 - n_2}{n_1 + n_2} \right) \right] \left[1 - \left(\frac{n_2 - n_3}{n_2 + n_3} \right) \left(\frac{n_3 - n_4}{n_3 + n_4} \right) \right] + \left[\left(\frac{n_0 - n_1}{n_0 + n_1} \right) - \left(\frac{n_1 - n_2}{n_1 + n_2} \right) \right] \left[\left(\frac{n_2 - n_3}{n_2 + n_3} \right) - \left(\frac{n_3 - n_4}{n_3 + n_4} \right) \right]}$$

$$R = |\Gamma|^2 = \left(\frac{\left[\left(\frac{n_0 - n_1}{n_0 + n_1} \right) - \left(\frac{n_1 - n_2}{n_1 + n_2} \right) \right] \left[1 - \left(\frac{n_2 - n_3}{n_2 + n_3} \right) \left(\frac{n_3 - n_4}{n_3 + n_4} \right) \right] + \left[1 - \left(\frac{n_0 - n_1}{n_0 + n_1} \right) \left(\frac{n_1 - n_2}{n_1 + n_2} \right) \right] \left[\left(\frac{n_2 - n_3}{n_2 + n_3} \right) - \left(\frac{n_3 - n_4}{n_3 + n_4} \right) \right]}{\left[1 - \left(\frac{n_0 - n_1}{n_0 + n_1} \right) \left(\frac{n_1 - n_2}{n_1 + n_2} \right) \right] \left[1 - \left(\frac{n_2 - n_3}{n_2 + n_3} \right) \left(\frac{n_3 - n_4}{n_3 + n_4} \right) \right] + \left[\left(\frac{n_0 - n_1}{n_0 + n_1} \right) - \left(\frac{n_1 - n_2}{n_1 + n_2} \right) \right] \left[\left(\frac{n_2 - n_3}{n_2 + n_3} \right) - \left(\frac{n_3 - n_4}{n_3 + n_4} \right) \right]} \right)^2$$

This equation represents the reflectivity R in terms of all the refractive indices, it is possible to simplify this equation even further, but it will be time consuming and does not serve any further purpose for this report. If the point was to optimize this formula to make our code run even faster, then we would need to simplify it.

B) Finding the refractive indices that minimize the reflectivity at the center wavelength

To find the relation between the refractive indices that minimize the reflectivity, we will set the Reflectivity to 0. To do that, we simply equate the numerator $T_{2,1}$ to 0.

$$(\Gamma_{01} - \Gamma_{12})(1 - \Gamma_{23}\Gamma_{34}) + (1 - \Gamma_{01}\Gamma_{12})(\Gamma_{23} - \Gamma_{34}) = 0$$

$$\left[\left(\frac{n_0 - n_1}{n_0 + n_1}\right) - \left(\frac{n_1 - n_2}{n_1 + n_2}\right)\right] \left[1 - \left(\frac{n_2 - n_3}{n_2 + n_3}\right) \left(\frac{n_3 - n_4}{n_3 + n_4}\right)\right] + \left[1 - \left(\frac{n_0 - n_1}{n_0 + n_1}\right) \left(\frac{n_1 - n_2}{n_1 + n_2}\right)\right] \left[\left(\frac{n_2 - n_3}{n_2 + n_3}\right) - \left(\frac{n_3 - n_4}{n_3 + n_4}\right)\right] = 0$$

After extensive algebraic manipulation and simplification, the equation reduces to the following form.

$$n_0(n_2)^2n_4 - (n_1)^2(n_3)^2 + n_0(n_2)^2n_4 - (n_1)^2(n_3)^2 = 0$$

$$n_0(n_2)^2n_4 = (n_1)^2(n_3)^2$$

Finally, we can obtain the relation

$$n_2 = \frac{n_1 n_3}{\sqrt{n_0 n_4}}$$

In this equation we assume $n_0=1$ and $n_4=3.5$. therefore, if we are given two of the refractive indices, we can easily solve for the third one and find a value that minimizes the reflectivity.

C) Minimizing the Reflectivity by finding n_2 for a given n_1 and n_3

Suppose we try to minimize the reflectivity of our system with a triple-layer antireflection coating, we are given that $n_1=1.4$ and $n_3=3.15$. From the nature of our system, we also know n_0 is 1 because we assume the incident wave travels in air. Lastly $n_4=n_{\text{cell}}=3.5$. With this data we can find the value of n_2 that will lead to a reflectivity of 0 at the center wavelength.

$$n_2 = \frac{n_1 n_3}{\sqrt{n_0 n_4}}$$

$$n_2 = \frac{1.4 * 3.15}{\sqrt{3.5}} = 2.36$$

In real life applications, if we wanted to find such a value for our second layer, we could fabricate it from zinc oxide (ZnO) which has a refractive index of 2.4 and is relatively safe and inexpensive⁴.

The Triple-Layer Coating Part 4

A) Finding the power for all values of n_2 between 1.4 and 3

By keeping the same refractive indices as in part 3 ; $n_0=1$, $n_1=1.4$, $n_3=3.15$ and $n_4=3.5$ we will calculate the power for every n_2 value between 1.4 and 3. Doing so, we will be able to verify the calculations we made in part 3 and validate if our equation to minimize the reflectivity also maximizes the power transmitted for the given n_2 . We first created a MATLAB algorithm that would calculate the total power for a spectrum for fixed refractive indices. Once this part worked successfully, we ran a loop that would calculate all the power values for $n_2=1.4$ to 3 in increments of 0.01. We tabulated all these results in APPENDIX I and we were able to plot them as well.

⁴ <https://www.sciencedirect.com/topics/chemistry/zinc-oxide-nanoparticle#:~:text=1.2.&text=Zinc%20oxide%20nanoparticle%20is%20the,et%20al.%2C%202018>

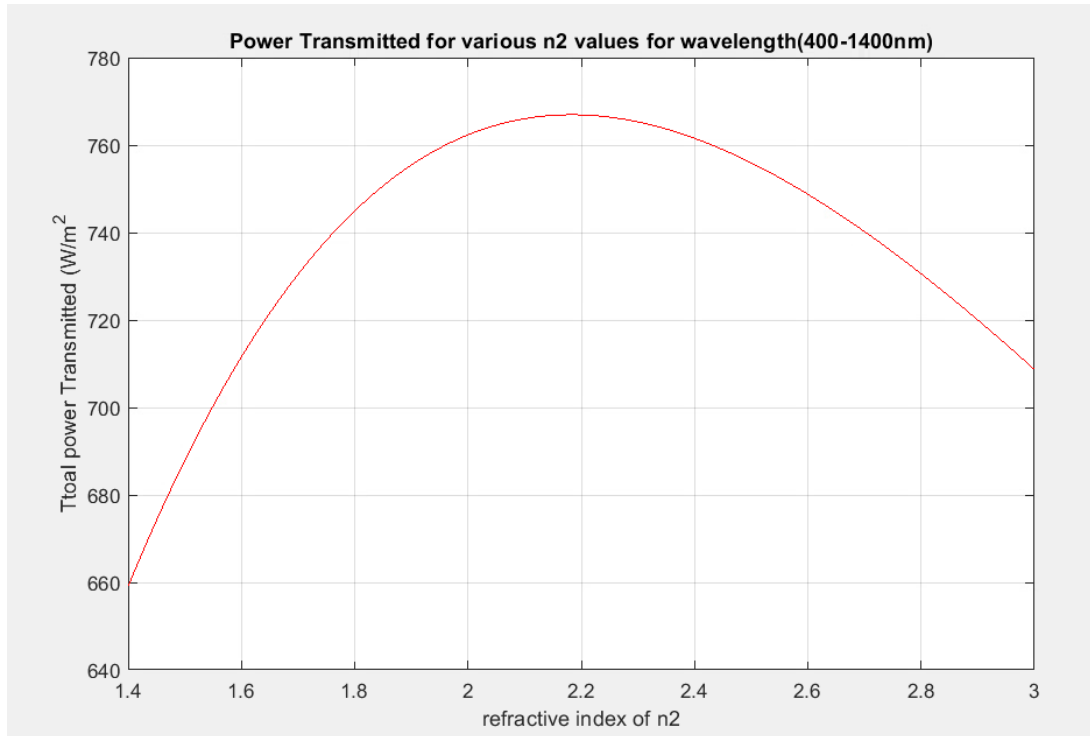


Figure 10 Total Power for different n_2 values for a wavelength (400nm-1400nm)

Looking at this graph and our tabulated results from APPENDIX I, for set values $n_1=1.4$ and $n_3=3.15$, we see that the max power transmitted of 766.9486 W/m^2 is found when $n_2= 2.18$.

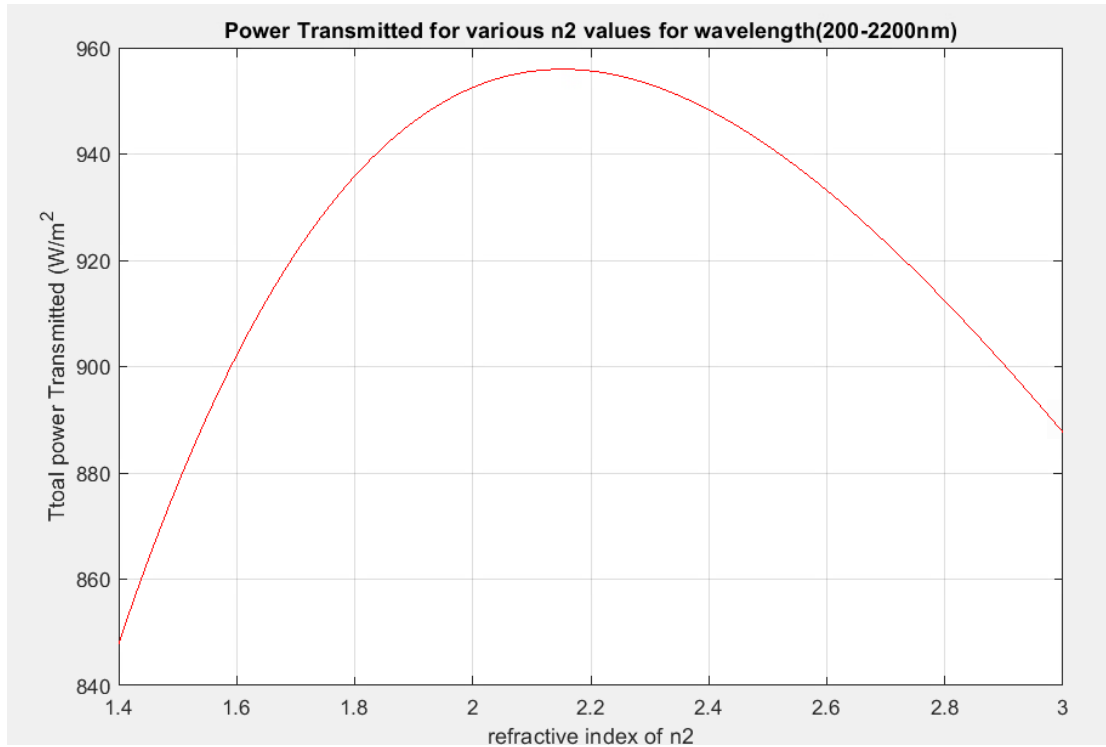


Figure 11 Total Power for different n_2 values for a wavelength (200nm-2200nm)

By plotting the total power transmitted for all n_2 values and fixed $n_1=1.4$ and $n_3=3.15$, for the wavelengths 200-2200 nm, we see that our most efficient value of $955.9468 W/m^2$ is found when n_2 is 2.15.

B) Determining the values that maximize the power

We can easily extract the maximum power obtained for each spectrum in our Result spreadsheet. We find that for the wavelength spectrum from 400 to 1400nm the best refractive index for n_2 is 2.18 allowing us to observe a total power of $766.9486 W/m^2$ and for the spectrum from 200 to 2200nm, the maximum power of $955.9468 W/m^2$ is obtained when $n_2=2.15$.

C) Evaluating the consistency of our analytical approach

The value obtained in the analytical approach was 2.36 for n_2 and this value disregarded the wavelength spectrum. When considering wavelengths from 200 to 2200 nm, we found that the most optimal n_2 was 2.15 and for a spectrum from 400 to 1400 nm we obtained $n_2=2.18$. Therefore, although the results we found in MATLAB are not too far from the one found in the analytical approach, we can't say that our values are consistent for both methods. Some of the

reasons for that are that the analytical approach was computed using the center frequency while the MATLAB approach utilized the entire spectrum from 400 to 1400nm and 220 to 2200nm. Being minimizing the reflectivity for the center wavelength only, we don't consider the impact other wavelengths could have on the total power for different wavelengths and different refractive indices. We know for a fact however that the cause of these inconsistencies is not because MATLAB rounds up the values to much considering that the software uses 15 to 16 significant digits⁵ in its calculations which is far beyond the precision we need for this project.

⁵ <https://www.mathworks.com/help/symbolic/digits.html>

Conclusion

This project aimed to demonstrate that the “simple” approach some may use to develop anti reflection coatings for solar cells by simply minimizing building their coatings layers whose refractive indices will minimize the reflection at the central wavelength may not always be the most optimal solution to maximize the power transferred to the cell. To prove this, made numerically integrated *Equation 2* using the TMM in order to determine the power transferred to the cell. Our strategy was to find the refractive indices that would minimize the reflectivity at the center wavelength of the system and then we would test every refractive index combination with the MATLAB software in hope of finding a better alternative.

We validated our work by computing the power for refractive indices that minimized the reflectivity at the center wavelength for both the double and triple-layer coatings. Luckily however, we did not attempt to maximize the overall power by testing each possibility for all three layers in the triple layer as the computational time would have been monstrous on my older computer. In fact, running my search algorithm that would find every possible solution for the double-layer coating took around 5 minutes. This would be a very interesting multi-disciplinary project I would like to make using the Golang programming language that is specifically made for concurrent processing which would without a doubt allow us to find an optimal solution for 3 layers in a much more reasonable period.

We graphed the reflectivity and as a function of the wavelength for both the double and triple layer coating. We graphed these values for two different spectrums: 400nm to 1400nm and 200nm to 2200nm. Surprisingly, by doubling the length of our spectrum, our total power only increased from 766.94W/m^2 to 955.94W/m^2 . As mentioned in the initial solar cell theory, this is because solar cells need a minimal energy to excite the electrons and produce electricity. For silicon cells, this minimal energy is generally attained for wavelength of 1100nm and lower⁶. Therefore, expanding the range from 1400nm to 2200nm had little on the total power. So, most of the extra power we measured came from the 200nm to 400nm range.

Finally, we computed the graph of the maximum power versus the refractive index of n_2 for the triple layer coating in order to verify if the optimal power value we found by selecting refractive indices values that minimized the reflectivity at the center wavelength would correspond to the n_2 value we find on the peak of our *Figures 9 and 10*. As expected, the values were not the same and we theorized that this is likely due to the fact that we are not using the same wavelength range for both of our calculations.

Overall, this project opened up the intricate and fascinating field of photovoltaic energy and the optical principles that surround anti-reflection coatings, which are crucial for enhancing solar cell efficiency. We made key assumptions that not only simplified the computational and

⁶ <https://sinovoltaics.com/learning-center/solar-cells/energy-band-gap-of-solar-cells/#:~:text=Silicon%2C%20the%20most%20popular%20semiconductor,energy%20greater%20than%201.11%20ev.>

mathematical challenges we faced but also allowed us to break down a complex analysis into more accessible and understandable parts. While this project only scratched the surface of this vast field, I found the concepts and science deeply engaging, and I definitely look forward to the opportunity to explore this technology further through future research.

Appendices

Appendix1

Part 4 A Data results for n2 maximization of power

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q
1																	
2																	
3	Project excel data collection part 4 a)							400-1400nm									200-2200nm
4	n2	Total pow	n2	Total pow	n2	total pow	n2	total pow		n2	Total pow	n2	Total pow	n2	total pow	n2	total pow
5	1.4	659.1	1.8	744.947	2.2	766.9138	2.6	748.753		1.4	847.8815	1.8	935.8444	2.2	955.6326	2.6	933.1471
6	1.41	662.2192	1.81	746.1697	2.21	766.8591	2.61	747.9567		1.41	851.1371	1.81	937.0523	2.21	955.4902	2.61	932.2254
7	1.42	665.2846	1.82	747.3528	2.22	766.78	2.62	747.1472		1.42	854.3333	1.82	938.218	2.22	955.3221	2.62	931.2898
8	1.43	668.2962	1.83	748.4968	2.23	766.6769	2.63	746.3249		1.43	857.4705	1.83	939.3422	2.23	955.1287	2.63	930.3407
9	1.44	671.2544	1.84	749.602	2.24	766.55	2.64	745.4898		1.44	860.5491	1.84	940.4252	2.24	954.9103	2.64	929.3783
10	1.45	674.1595	1.85	750.6689	2.25	766.3997	2.65	744.6424		1.45	863.5695	1.85	941.4675	2.25	954.6673	2.65	928.4028
11	1.46	677.0117	1.86	751.6979	2.26	766.2263	2.66	743.7827		1.46	866.5319	1.86	942.4696	2.26	954.4001	2.66	927.4144
12	1.47	679.8114	1.87	752.6893	2.27	766.0301	2.67	742.911		1.47	869.4369	1.87	943.4318	2.27	954.1089	2.67	926.4135
13	1.48	682.5587	1.88	753.6436	2.28	765.8115	2.68	742.0275		1.48	872.2848	1.88	944.3546	2.28	953.7941	2.68	925.4001
14	1.49	685.2541	1.89	754.5611	2.29	765.5708	2.69	741.1324		1.49	875.076	1.89	945.2385	2.29	953.4561	2.69	924.3746
15	1.5	687.8979	1.9	755.4423	2.3	765.3083	2.7	740.226		1.5	877.811	1.9	946.0838	2.3	953.0951	2.7	923.3371
16	1.51	690.4903	1.91	756.2876	2.31	765.0243	2.71	739.3084		1.51	880.4901	1.91	946.8911	2.31	952.7116	2.71	922.2879
17	1.52	693.0317	1.92	757.0973	2.32	764.7192	2.72	738.3799		1.52	883.1138	1.92	947.6608	2.32	952.3058	2.72	921.2271
18	1.53	695.5224	1.93	757.872	2.33	764.3932	2.73	737.4406		1.53	885.6824	1.93	948.3933	2.33	951.8781	2.73	920.1551
19	1.54	697.9628	1.94	758.6118	2.34	764.0466	2.74	736.4907		1.54	888.1965	1.94	949.089	2.34	951.4288	2.74	919.0719
20	1.55	700.3532	1.95	759.3174	2.35	763.6797	2.75	735.5305		1.55	890.6564	1.95	949.7484	2.35	950.9582	2.75	917.9779
21	1.56	702.6939	1.96	759.9889	2.36	763.2929	2.76	734.5601		1.56	893.0626	1.96	950.3719	2.36	950.4666	2.76	916.8731
22	1.57	704.9853	1.97	760.6269	2.37	762.8864	2.77	733.5797		1.57	895.4155	1.97	950.9599	2.37	949.9544	2.77	915.7579
23	1.58	707.2278	1.98	761.2318	2.38	762.4606	2.78	732.5896		1.58	897.7155	1.98	951.5128	2.38	949.4218	2.78	914.6324
24	1.59	709.4217	1.99	761.8039	2.39	762.0157	2.79	731.5899		1.59	899.963	1.99	952.0311	2.39	948.8692	2.79	913.4968
25	1.6	711.5674	2	762.3436	2.4	761.5519	2.8	730.5808		1.6	902.1586	2	952.5151	2.4	948.2968	2.8	912.3513
26	1.61	713.6653	2.01	762.8512	2.41	761.0697	2.81	729.5625		1.61	904.3027	2.01	952.9654	2.41	947.705	2.81	911.1961
27	1.62	715.7157	2.02	763.3273	2.42	760.5693	2.82	728.5351		1.62	906.3956	2.02	953.3823	2.42	947.0941	2.82	910.0314
28	1.63	717.719	2.03	763.7721	2.43	760.0509	2.83	727.4989		1.63	908.4379	2.03	953.7662	2.43	946.4643	2.83	908.8573
29	1.64	719.6756	2.04	764.186	2.44	759.5149	2.84	726.454		1.64	910.43	2.04	954.1175	2.44	945.816	2.84	907.6741
30	1.65	721.5858	2.05	764.5695	2.45	758.9615	2.85	725.4006		1.65	912.3723	2.05	954.4367	2.45	945.1494	2.85	906.482
31	1.66	723.4501	2.06	764.9229	2.46	758.391	2.86	724.3388		1.66	914.2654	2.06	954.7241	2.46	944.4648	2.86	905.2811
32	1.67	725.2689	2.07	765.2465	2.47	757.8036	2.87	723.2689		1.67	916.1095	2.07	954.9801	2.47	943.7626	2.87	904.0715
33	1.68	727.0425	2.08	765.5407	2.48	757.1997	2.88	722.191		1.68	917.9053	2.08	955.2052	2.48	943.0429	2.88	902.8536
34	1.69	728.7713	2.09	765.806	2.49	756.5794	2.89	721.1053		1.69	919.6531	2.09	955.3997	2.49	942.3061	2.89	901.6274
35	1.7	730.4557	2.1	766.0427	2.5	755.9431	2.9	720.012		1.7	921.3535	2.1	955.5641	2.5	941.5525	2.9	900.3931
36	1.71	732.0961	2.11	766.251	2.51	755.291	2.91	718.9111		1.71	923.0067	2.11	955.6986	2.51	940.7822	2.91	899.151
37	1.72	733.693	2.12	766.4315	2.52	754.6234	2.92	717.8029		1.72	924.6134	2.12	955.8038	2.52	939.9957	2.92	897.9011
38	1.73	735.2467	2.13	766.5845	2.53	753.9405	2.93	716.6875		1.73	926.174	2.13	955.88	2.53	939.1931	2.93	896.6436
39	1.74	736.7576	2.14	766.7102	2.54	753.2426	2.94	715.5651		1.74	927.6889	2.14	955.9275	2.54	938.3747	2.94	895.3787
40	1.75	738.2261	2.15	766.8092	2.55	752.5298	2.95	714.4358		1.75	929.1586	2.15	955.9468	2.55	937.5407	2.95	894.1065
41	1.76	739.6526	2.16	766.8816	2.56	751.8026	2.96	713.2998		1.76	930.5835	2.16	955.9382	2.56	936.6915	2.96	892.8273
42	1.77	741.0376	2.17	766.928	2.57	751.061	2.97	712.1572		1.77	931.964	2.17	955.9021	2.57	935.8273	2.97	891.5411
43	1.78	742.3814	2.18	766.9486	2.58	750.3054	2.98	711.0083		1.78	933.3008	2.18	955.8389	2.58	934.9484	2.98	890.2482
44	1.79	743.6844	2.19	766.9437	2.59	749.536	2.99	709.853		1.79	934.5941	2.19	955.7489	2.59	934.0549	2.99	888.9486
45							3	708.6916								3	887.6425
46																	

References

- E. Gad, “ELG3136 ANALOG ELECTRONICS II” University of Ottawa
- H. Schriemer, “ELG 3106 2024 Project”, University of Ottawa
- J.Svarc, “Most efficient solar panels 2024”, [Online article], Clean Energy Reviews, <https://www.cleanenergyreviews.info/blog/most-efficient-solar-panels>
- SciToons, «Dow do solar cells work?», [online Video], <https://www.youtube.com/watch?v=UJ8XW9AgUrw&t=304s>
- N. D. Weimar “Energy Band gap Solar cells” , [Online article], Sino Voltaics, <https://sinovoltaics.com/learning-center/solar-cells/energy-band-gap-of-solar-cells/#:~:text=Silicon%2C%20the%20most%20popular%20semiconductor,energy%20gr eater%20than%201.11%20ev.>
- KMLABS Inc, “Photon Unit Calculator”, <https://www.kmlabs.com/en/wavelength-to-photon-energy-calculator>
- <https://www.mathworks.com/help/symbolic/digits.html>
- T. Lakshmipriya, S. Gopinath, “Introduction to nanoparticles and analytical devices”, [Online article], Science Direct, <https://www.sciencedirect.com/topics/chemistry/zinc-oxide-nanoparticle#:~:text=1.2.&text=Zinc%20oxide%20nanoparticle%20is%20the,et%20al.%2C%202018>